

Homework — 1.2.4 Types of Programming Language — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] A programming paradigm is a style/approach to writing programs, e.g. procedural, object-oriented, assembly (1). Different problems suit different paradigms — some problems are easier to express/maintain/reuse in one paradigm than another, so the programmer chooses the paradigm that fits the task (1).
2. [2] Any two (1 each): program is built from procedures/subroutines that can be called; uses the structured constructs sequence, selection and iteration; instructions are executed in a defined step-by-step order (states *how* the task is done); uses variables/data structures to hold state. Max 2.
3. (a) [2] Output is **9** (1). The program outputs the **larger** of the two numbers input (accept: if they are equal it outputs that value) (1).
- (b) [2] BRP branches to (sets the PC to) the address labelled **big** **only if the Accumulator is positive or zero** (1). It is needed to test whether $\text{second} - \text{first} \geq 0$, i.e. to select which of the two numbers is output / to implement the selection (1).
4. [4] (a) Immediate = **4** — the operand itself is the value (1). (b) Direct = **7** — the value held at address 4 (1). (c) Indirect = **2** — address 4 holds 7, which is used as the address; address 7 holds 2 (1). (d) Indexed = **9** — effective address = operand 4 + index register 2 = address 6, which holds 9 (1).
5. [2] Inheritance: a subclass acquires (inherits) the attributes and methods of a superclass, and may add its own or override them (1). Benefit (1, any one): avoids duplicating code / promotes reuse; makes the program easier to maintain and extend; changes to shared behaviour are made once in the superclass. Max 2.

Part B (6 marks)

6. [3] Advantages (1 each, max 2): simple to manage/plan as stages are completed in sequence; well documented at each stage; clear milestones/deliverables. Disadvantage (1, any one): inflexible — difficult and costly to return to an earlier completed stage; the client only sees the product at the end, so requirement errors are found late. Max 3.
7. [2] Static linking copies the required library code into the executable at compile/link time, so the executable is self-contained but larger, and library updates require re-linking (1). Dynamic linking keeps the library in a separate file (e.g. a DLL) that is linked/loaded at run time, so the executable is smaller and the library can be shared and updated independently (1).
8. [1] The Accumulator temporarily holds the data being processed / the results of arithmetic and logic operations carried out by the ALU (1).

Total: 14 + 6 = 20